

CAREERTOURS LLM - 40 COURSE KNOWLEDGE, ASSESSMENT & CAREER CORPUS

Approved Knowledge Corpus v3.0 | Minimum 10 substantive pages per course

Purpose: Provide a versioned, evidence-based knowledge corpus for CareerTours retrieval, assessment generation, remediation and career matching. This document expands the approved Version 2.1 specification without changing the 40 course codes, syllabus order or compliance rules.

Human-readable source	This PDF supports trainers, developers, QA and management review.
Machine-readable source	The companion JSONL contains atomic records for ingestion, retrieval, evaluation and controlled fine-tuning.
Recommended architecture	Use retrieval-augmented generation first. Fine-tune only after validated interaction data and evaluation evidence justify it.
Data boundary	No real student records, phone numbers, resumes or personal data are included in this corpus.
Release gate	The corpus does not replace trainer approval of live scored questions or course-specific operational availability.

Do not upload the PDF alone and assume the model is trained. Chunk the JSONL by record type, index approved records with metadata, retrieve relevant evidence at runtime and evaluate answers against a locked test set.

Recommended Model Architecture

RAG first; fine-tuning only for stable behaviour

Area	Approved requirement
1. Source layer	Approved PDF, JSONL records and primary references; immutable version IDs.
2. Ingestion	Validate schema, content hash, status, effective date and course/section relationships.
3. Retrieval	Filter by course_code, section_id, language, version, record_type and approval status before semantic ranking.
4. Generation	Supply only the necessary approved chunks plus response schema and safety rules.
5. Validation	Check citations, allowed claims, IDs, score calculations and schema before showing output.
6. Evaluation	Run locked course, assessment, safety, privacy and career-match test sets before every release.
7. Fine-tuning	Use only de-identified, reviewed examples when prompt/RAG improvements are insufficient.

All production answers must remain grounded in approved records and visible evidence. Model fluency is not proof of correctness.

Corpus Record Types

Atomic units make retrieval and maintenance reliable

Area	Approved requirement
course_profile	Identity, learner, prerequisite, duration, outcome, tools and boundaries.
module	Concepts, objectives, knowledge statements, application, evidence and misconceptions.
section	Ordered modules, skill tags, completion evidence and assessment blueprint.
assessment_sample	Non-live sample question, answer, explanation, rubric and difficulty.
project	Brief, deliverables, skills, rubric and verification.
occupation_mapping	Role title, weighted skills, critical skills, evidence minimum and gap actions.
faq / misconception	Approved response plus retrieval and escalation metadata.
guardrail / policy	Privacy, safety, compliance, versioning and prohibited claims.

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Retrieval and Chunking Rules

Keep facts attributable and context small

Area	Approved requirement
Chunk unit	One atomic JSONL record; do not split an answer from its course, section, version or evidence.
Mandatory filters	approved status, active corpus version, course_code and relevant record_type.
Ranking	Metadata filter first, then semantic relevance; boost exact section and skill-tag matches.
Context budget	Retrieve the smallest sufficient evidence set; remove near-duplicates.
Citation	Return record_id and source_version for every material course, assessment or career claim.
Conflict	Prefer the newest effective approved record; surface unresolved conflicts instead of blending them.
No answer	When approved evidence is insufficient, say what is missing and request trainer/developer review.

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Fine-Tuning Data Policy

What may and may not be used

Area	Approved requirement
Allowed later	Reviewed, de-identified question classifications, remediation examples, response formatting and role-explanation examples.
Not allowed	Raw resumes, names, phone numbers, chats, Aadhaar/PAN/bank data, private trainer notes or unreviewed model outputs.
Training split	Separate train, validation and locked test sets by student and content family to prevent leakage.
Deduplication	Remove exact and semantic duplicates before splitting.
Provenance	Every example keeps source_record_ids, reviewer, consent basis, model/prompt version and date.
Rollback	Retain dataset and model versions; a release must be reversible.
Decision rule	Do not fine-tune merely because data exists; prove improvement on the locked evaluation suite.

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Evaluation Suite

Measure usefulness, truthfulness and safety

Area	Approved requirement
Course accuracy	Correct modules, order, prerequisites, tools, evidence and boundaries.
Section boundary	No question or explanation introduces later/unapproved topics.
Assessment quality	Unambiguous answer, aligned difficulty, valid rubric and useful explanation.
Career consistency	Deterministic weights, gates, confidence and evidence citations.
Remediation	Specific section, skill, module, practice task and retest condition.
Privacy	No unnecessary personal-data request or exposure.
Safety/compliance	No job guarantee, unsafe cyber instruction, fabricated law/rate/policy or hidden assumption.
Robustness	Handles missing, conflicting, stale and adversarial input without inventing an answer.

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Human Review and Release Governance

People remain accountable for high-impact content

Area	Approved requirement
Trainer	Technical accuracy, learning sequence, practical feasibility and course evidence.
Assessment owner	Blueprint, difficulty, ambiguity, rubrics, duplicates, accessibility and appeals.
Career reviewer	Role title, skill weights, evidence minimums and gap actions.
Privacy/security	Data minimisation, access, retention, provider processing and incident controls.
Release owner	Approves corpus/model/prompt combination after UAT and evaluation.
Audit	Record author, reviewer, timestamp, reason, previous version and affected results.
Emergency stop	Disable a record, course or model route without deleting history.

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Provider-Neutral LLM Contract

Do not lock CareerTours to one model

Area	Approved requirement
Input	task, course_code, section_id if applicable, retrieved record_ids, language and output schema.
Output	structured JSON plus record citations; no hidden score calculations.
Failure	Malformed output, timeout or missing evidence returns recoverable status; never fabricate a result.
Observability	Log provider/model, prompt version, retrieved record IDs, latency, token/cost metrics and validation outcome.
Fallback	Switch models only through the same schema, tests and safety validation.
Secrets	Never place API keys or credentials in prompts, logs or course records.
Chain of thought	Do not request or expose hidden reasoning; provide concise evidence-based explanations.

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Multilingual and Accessibility Standard

English and Telugu support without changing meaning

Area	Approved requirement
Canonical content	English remains the scored canonical version for technical terms, IDs, answers and rubrics.
Telugu support	Provide a separate explanation layer; do not change the competency, marks or correct answer.
Glossary	Keep technical terms in English with clear Telugu/plain-language explanation where helpful.
Accessibility	Keyboard navigation, labelled controls, readable contrast, resizable text, screen-reader semantics and extra-time accommodation.
Low bandwidth	Autosave, resumable attempts and text-first alternatives where the skill does not require media.
Quality check	Bilingual content requires human review for meaning, not only literal translation.

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Data Privacy and Security

Minimum data, minimum access, maximum auditability

Area	Approved requirement
Registration	Student name and phone only.
Resume	Optional, consented and minimised before external processing.
Storage	Encryption, malware scanning, role-based access, backups and tested restoration.
Logs	No resume body, question answers, secrets or unnecessary personal data in application logs.
Retention	Nipuna-approved schedule plus correction, export and deletion workflow.
Test data	Use synthetic records; never copy production student data into development.
Incident	Detect, contain, notify, correct and document according to the approved response plan.

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Versioning and Change Control

Every answer must know which knowledge version it used

Area	Approved requirement
Version units	Corpus, course, module, question, role mapping, prompt, model and scoring formula.
Effective dates	Future versions may be approved before activation; only one active version per route.
Student attempt	Freeze the syllabus, question set and score formula used for that attempt.
Correction	Create a new record; rescore consistently when a defect materially affected students.
Deprecation	Retired records remain queryable for audit but cannot enter new production outputs.
Release notes	State added, changed, removed, migration impact and reviewer.

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Implementation Sequence

Recommended delivery to reduce risk

Area	Approved requirement
Phase 1	Ingest course_profile, module, section, project and policy records; verify retrieval citations.
Phase 2	Load reviewed sample assessments and trainer-approved item bank; implement scoring and remediation.
Phase 3	Implement role taxonomy, confidence and explainable top-three matches.
Phase 4	Pilot five categories, inspect errors and improve corpus/prompts.
Phase 5	Release all 40 courses in controlled batches with monitoring.
Phase 6	Consider de-identified fine-tuning only after RAG performance and failure evidence are measured.

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